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Contributors

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Correctly rounded floating-point maths functions

1. Motivation

Floating-point maths functions are rounded after calculation. The rounding mode used is part of the floating-point state which is maintained per-thread. This introduces two problems:

1. Changing rounding modes, for example for calculations that require correct rounding in a set of optimised expression evaluations, is unergonomic.
2. There is a burden on library writers to restore the floating-point state to the condition it was in when a library function was called.

This is not a new problem, and the C standard in Annex F reserves the prefix `cr_` for functions fully matching the IEC 60559 mathematical operations. This paper proposes adding five overload sets to the standard library for addition, subtraction, multiplication, division and square root calculation.

These functions guarantee that the operation will be carried out as if with infinite precision, and rounded using the round-nearest-to-even rounding mode, as specified in [60559:2020](#). The parameter type must satisfy the trait `std::numeric_limits<T>::is_iec559`.

Other functions, such as trig functions and complex maths functions, are not available for correctly rounded implementations. This would require specifying an algorithm, which is outside the scope of this standard. It is hoped that cross-implementation third-party libraries will be developed which will offer solutions to this problem.

This solves problem 1 directly by providing a more ergonomic way of expressing inten-